

19) $\int_{-1}^1 \left(\underbrace{x^3 \cos x}_{\substack{\text{Tek} \\ \text{Fonk.} \\ \text{0}}}} + \underbrace{\sin^7 x}_{\substack{\text{Tek} \\ \text{f.} \\ \text{0}}} - \underbrace{x^2 \sin x}_{\substack{\text{Tek} \\ \text{fonk.} \\ \text{0}}} + \underbrace{x^4}_{\substack{\text{Çift} \\ \text{Fonk.}}} \right) dx = ?$ A) $\frac{\cos 1}{3} + \frac{4}{5}$ B) $\frac{2}{5}$ C) $\frac{\cos 1}{3} - \frac{4}{5}$ D) $\frac{3}{5}$ E) 0

$\int_{-1}^1 (x^3 \cos x + \sin^7 x - x^2 \sin x + x^4) dx = \int_{-1}^1 x^4 dx = \frac{x^5}{5} \Big|_{-1}^1 = \frac{2}{5}$

$\int \left(\frac{2x}{x^2+3} - e^{5x} + \sin 4x - \frac{1}{x^2+9} \right) dx = ?$

A) $\ln(x^2+3) - \frac{e^{5x}}{5} - \frac{\cos 4x}{4} - \frac{1}{3} \arctan \frac{x}{3} + c$

B) $\frac{1}{\sqrt{3}} \arctan \frac{x}{\sqrt{3}} - \frac{e^{5x}}{5} + 4 \cos 4x - \ln(x^2+9) + c$

C) $\ln(x^2+3) - \frac{e^{5x}}{5} - 4 \cos 4x - \arctan \frac{x}{3} + c$

D) $\frac{1}{\sqrt{3}} \arctan \frac{x}{\sqrt{3}} - 5e^{5x} + 4 \cos 4x - \ln(x^2+9) + c$

E) $\ln(x^2+3) - 5e^{5x} - \frac{\cos 4x}{4} - \arctan \frac{x}{3} + c$

20) $\lim_{x \rightarrow 0} \frac{\int_0^{x^2} \sin(t^2) dt}{2x^6} = ?$ A) ∞ B) 0 C) $\frac{1}{4}$ D) $\frac{1}{2}$ E) $\frac{1}{6}$

$\hookrightarrow \frac{0}{0} \rightarrow L'H.$

$\lim_{x \rightarrow 0} \frac{2x \cdot \sin(x^2)^2 - 0}{12x^5} = \lim_{x \rightarrow 0} \frac{\sin x^4}{6x^4} = \frac{1}{6}$

21) $\int \frac{dt}{t^2(t^4-16)}$ integralinin basit kesirlere ayrılmış hali $\rightarrow t^2 \cdot (t-2)(t+2)(t^2+4)$ aşağıdakilerden hangisi olabilir?

$\rightarrow 5$ çarpan var 5 kesir olmalı

A) $\int \left(\frac{A}{t} + \frac{B}{t^2} + \frac{C}{t+2} + \frac{D}{(t-2)^2} + \frac{Et+F}{t^2+4} \right) dt$

B) $\int \left(\frac{A}{t} + \frac{B}{t-2} + \frac{C}{t+2} + \frac{Dt+E}{t^2+4} \right) dt$

C) $\int \left(\frac{A}{t} + \frac{B}{t-2} + \frac{Ct+D}{(t-2)^2} + \frac{E}{t+2} + \frac{Ft+G}{t^2+4} \right) dt$

D) $\int \left(\frac{A}{t} + \frac{B}{t^2} + \frac{C}{t-2} + \frac{D}{t+2} + \frac{Et+F}{t^2+4} \right) dt$

E) $\int \left(\frac{A}{t} + \frac{B}{t^2} + \frac{C}{t+2} + \frac{Dt+E}{t^2+4} \right) dt$

(22) $F(x) = \int_0^x \cos(\pi t^2) dt$ eğrisinin $x = \frac{1}{\sqrt{6}}$

noktasındaki teğetin eğimi aşağıdakilerden hangisidir?

$\Rightarrow F'(\frac{1}{\sqrt{6}})$ y' soruyor.

- A) $\frac{\sqrt{3}}{2}$ B) 1 C) $\frac{\pi}{\sqrt{3}} - 1$ D) $1 - \frac{1}{\sqrt{3}}$ E) $\sqrt{3} - 1$

$F'(x) = \cos \pi x^2 \rightarrow F'(\frac{1}{\sqrt{6}}) = \cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$

(23) $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\sin x)^3 dx = ?$

- A) 1 B) $\frac{\sqrt{3}}{8}$ C) $\frac{2}{3} - \frac{5\sqrt{2}}{12}$ D) $\frac{2}{3} - \frac{\sqrt{3}}{8}$ E) $\frac{5\sqrt{2}}{12}$

$\int_{\pi/4}^{\pi/2} \sin^2 x \cdot \sin x dx = \int_{\pi/4}^{\pi/2} (1 - \cos^2 x) \sin x dx$

$\cos x = u \quad - \sin x dx = du$

$x = \frac{\pi}{2} \rightarrow u = 0$

$x = \frac{\pi}{4} \rightarrow u = \frac{\sqrt{2}}{2}$

$= \int_{\sqrt{2}/2}^0 (1 - u^2) (-du) = \int_0^{\sqrt{2}/2} (1 - u^2) du = u - \frac{u^3}{3} \Big|_0^{\sqrt{2}/2} = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{12} = \frac{5\sqrt{2}}{12}$

(24) $\int_1^{e^{\pi/4}} \frac{\sec^2(\ln x)}{x} dx = ?$

$\ln x = u$

$\frac{dx}{x} = du$

$x = e^{\pi/4} \rightarrow u = \pi/4$

- A) $e^{\frac{\pi}{4}} - \ln \frac{\sqrt{2}}{2}$ B) 1 C) $\frac{\pi}{4} - 1$ D) $1 - \ln \sqrt{2}$ E) $\frac{\pi^3}{4^3} + \ln \sqrt{2}$

$x = 1 \rightarrow u = 0$

$\int_0^{\pi/4} \sec^2 u du = \tan u \Big|_0^{\pi/4} = 1$

(25) $\int \frac{x^3}{\sqrt{5+x^2}} dx$ integralini çözebilmek için yapılması gereken trigonometrik dönüşüm ve bu dönüşüm sonrası oluşan yeni integral aşağıdakilerin hangisinde doğru verilmiştir?

A) Yapılması gereken dönüşüm: $x = \sqrt{5} \tan t$, Yeni integral: $\int 5\sqrt{5}(\tan t)^3 \cdot (\sec t)^2 dt$

B) Yapılması gereken dönüşüm: $x = 5 \sin t$, Yeni integral: $\int \sqrt{5}(\sin t)^3 \cdot \cos t dt$

C) Yapılması gereken dönüşüm: $x = 5 \tan t$, Yeni integral: $\int (\tan t)^3 \cdot \sec t dt$

D) Yapılması gereken dönüşüm: $x = \sqrt{5} \sin t$, Yeni integral: $\int 5\sqrt{5}(\sin t)^3 dt$

(E) Yapılması gereken dönüşüm: $x = \sqrt{5} \tan t$, Yeni integral: $\int 5\sqrt{5}(\tan t)^3 \cdot \sec t dt$

$$x = \sqrt{5} \tan t \quad dx = \sqrt{5} \sec^2 t dt \quad \sqrt{5+x^2} = \sqrt{5+5\tan^2 t} = \sqrt{5} \sec t$$

$$\int \frac{x^3}{\sqrt{5+x^2}} dx = \int \frac{5\sqrt{5} \tan^3 t}{\sqrt{5} \sec t} \cdot \sqrt{5} \sec^2 t dt = \int 5\sqrt{5} \tan^3 t \sec t dt$$

(26) $\int_0^5 \operatorname{arccot} x dx = ?$ A) $25 \operatorname{arccot} 5$ B) $5 \operatorname{arccot} 5 + \ln 26$ C) $5 \operatorname{arccot} 5 + \frac{\ln 26}{2}$ D) $25 \operatorname{arccot} 5 + \frac{\ln 26}{2}$ E) $25 \operatorname{arccot} 5 + \ln 26$

Kısmi int.

$$u = \operatorname{Arccot} x \quad dv = - \frac{dx}{1+x^2}$$

$$dv = dx \quad v = x$$

$$\int_0^5 \operatorname{Arccot} x dx = x \operatorname{Arccot} x \Big|_0^5 + \int_0^5 \frac{x}{1+x^2} dx$$

$$\quad \quad \quad \frac{1}{2} \ln(1+x^2)$$

$$= 5 \operatorname{Arccot} 5 + \frac{1}{2} \ln(1+x^2) \Big|_0^5$$

$$= 5 \operatorname{Arccot} 5 + \frac{1}{2} \ln 26$$